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New Sickle Cell Therapies Will Be Out of Reach Where They Are Needed Most

There is no clear path for African patients to get access to the treatments, which have multimillion-dollar price tags and are highly complex to manufacture and deliver.




By Rebecca Robbins and Stephanie Nolen
The reporters interviewed sickle cell patients, doctors and researchers in seven countries.

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The Food and Drug Administration's approval on Friday of two groundbreaking gene therapy treatments for sickle cell disease has brought a rare moment of hope and celebration to people with the agonizing blood disorder.

But there is no clear path for the new therapies — one-time treatments so effective in clinical trials that they have been hailed as cures — to reach the countries where the vast majority of people with sickle cell live. Shortly after the approval their manufacturers announced sticker prices in the millions of dollars: $3.1 million for Lyfgenia, made by Bluebird Bio, and $2.2 million for Casgevy, made by Vertex Pharmaceuticals.

Lyfgenia will launch in the United States. Vertex has been prioritizing winning approval in six wealthy countries — the United States, Italy, Britain, France, Germany and Saudi Arabia — that, by one estimate, are home to 2 percent of the

global sickle cell population.

Three-quarters of the world's sickle cell patients are in sub-Saharan Africa. Several million of them are believed to be sick enough that they would be eligible for the new therapies, compared with some 20,000 in the United States.

Many African patients have been closely following news online of the treatments' success in clinical trials. In Tanzania, information about Casgevy spread a few months ago through a WhatsApp group that Shani Mgaraganza set up for mothers of children with sickle cell. Her son, Ramadhani, 12, and daughter Nasra, 10, have the inherited disorder, which causes episodes of searing pain and damages their organs. She said the therapy sounded like a miracle.

"Everyone said, 'Thanks God, our kids will be well,'" she said.

Then the mothers learned what it was likely to cost. "It would be billions of Tanzanian shillings," Ms. Mgaraganza said. "No one can afford this. It was demoralizing."

The access question is front of mind for Jennifer Doudna, the University of California, Berkeley, scientist who shared a Nobel Prize in Chemistry for pioneering the gene-editing method, CRISPR, underpinning Casgevy. "Today it's not going to be widely available," she said. "Now that we have this approval, we need to really figure out how we're going to open it up to more people."

Two key factors put it out of the reach of patients in Africa.

First, price: The treatments are far too expensive for governments that struggle to pay for basic health services. In some cases, there may be substantial additional costs, such as for a patient's extended hospital stay to receive gene therapy.

The second barrier is medical infrastructure: Administering the treatment is a monthslong process at medical centers that can perform stem cell transplants. Patients must have their cells harvested and flown to a lab for manufacturing, undergo grueling chemotherapy and endure a long hospitalization.

“A medicine that is so resource-intensive as this is may not be appropriate in many places where the amount of resources for health care is more limited,” said Vertex’s chief scientific officer, Dr. David Altshuler.



Casgevy is manufactured by Vertex Pharmaceuticals, which has been prioritizing winning approval in wealthy countries that are home to only a small fraction of the global sickle cell population. Vertex Pharmaceuticals



Dr. Obiageli Nnodu, director of the Centre of Excellence for Sickle Cell Disease Research and Training at the University of Abuja said the new treatments are a source of joy but “a moonshot away” for her patients. Taiwo Aina for The New York Times

He said the company was working to develop cheaper and easier approaches to treat sickle cell patients around the world, including a simple pill, which has not yet been tested in humans. “This takes a long time to do, and I feel like we’re at the beginning of the next phase,” he said.

A Bluebird Bio spokeswoman, Jess Rowlands, said it was “the unfortunate reality” that the infrastructure needed for such gene therapies “does not exist in much of the world.” Bluebird will “continue to invest in approaches that may support global access in the future,” she said.

New medicines often debut in wealthy countries years before they arrive in poorer parts of the world. The disparities have been exacerbated in recent years, as a wave of cutting-edge therapies with breathtaking price tags have transformed the lives of patients in wealthy countries. Vertex in particular has faced criticism for keeping its pioneering cystic fibrosis drugs out of reach of thousands of patients in lower-income countries.

Manufacturers charge high prices even when it is clear that lower-income countries can’t afford to pay those costs to protect their ability to demand higher prices in places like the United States and Europe.

Dr. Obiageli Nnodu, director of a sickle cell program at the University of Abuja in Nigeria, has discussed the new gene therapy with some of her patients. It’s a source of joy, she said, “but it’s a moonshot away.”

The mutation that causes sickle cell is thought to have arisen 7,000 years ago in West Africa. (Worldwide, most people with the disease are of African ancestry.) It became most common in places where malaria was endemic because a single copy of the gene protects against infection with malaria. But two copies of the gene cause red blood cells to deform in the shape of a sickle that can block blood vessels, causing excruciating pain, strokes and other problems that shorten lives.



Stephanie Nolen
Global health reporter covering access to medicines and health care

“The guiding principle of my work is ‘go there.’ I want to hear directly from the people who are affected by disease, or lack of access to a new drug. I’ve been writing about global health for 30 years and have reported from more than 80 countries.”

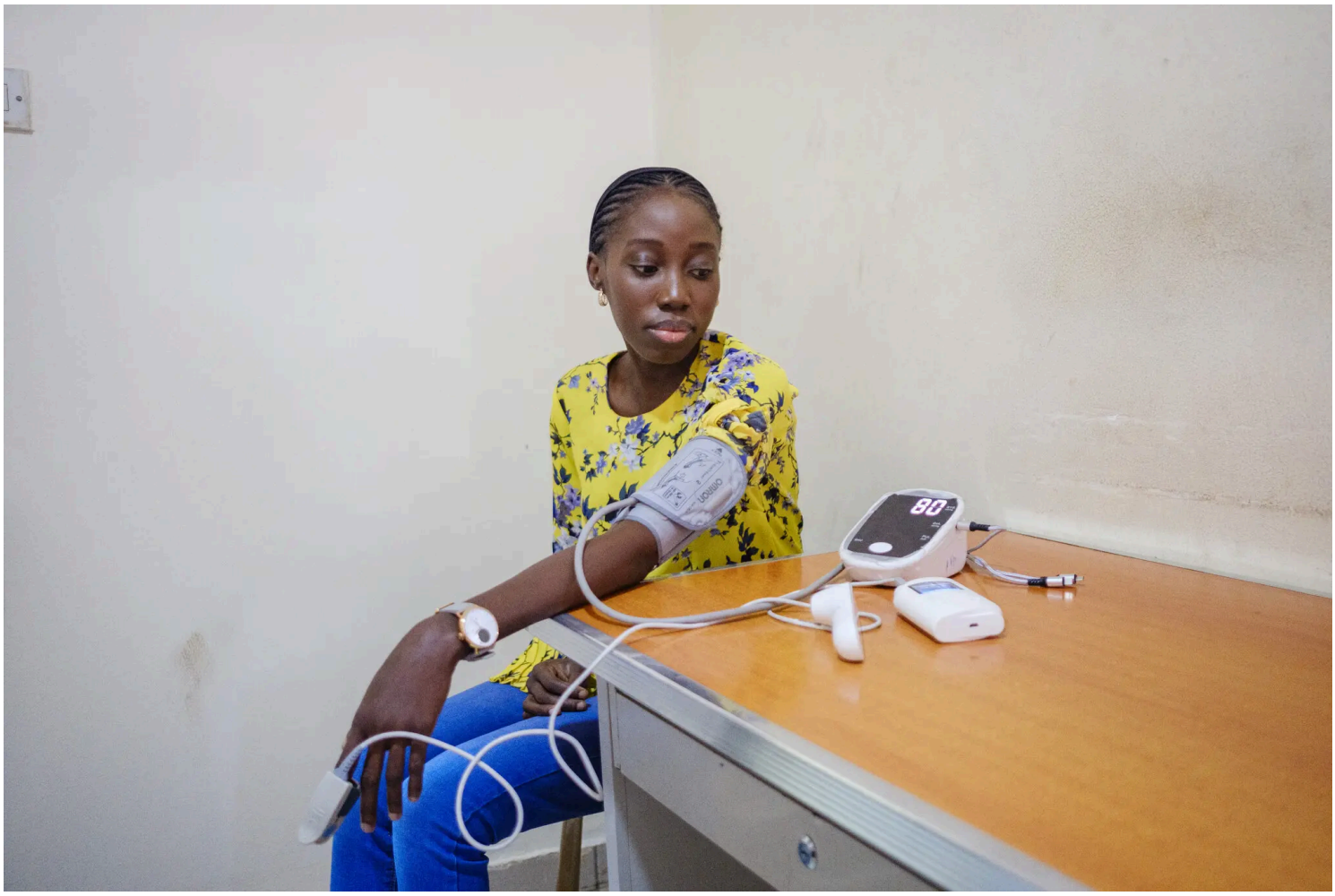
Learn more about how Stephanie Nolen approaches her work.

Casgevy modifies patients' DNA to correct the underlying cause of their illness. In a clinical trial, it eliminated pain crises for 29 out of 31 patients with sickle cell.

Questions about access extend to the United States, where many people with the disease are poor and live in states that have not expanded their Medicaid programs.

Those who can get Casgevy in the United States and other high-income countries will go to medical centers accredited to perform stem cell transplants. Nearly 200 clinical programs in eight countries — the vast majority in the United States and none in Africa — have received that stamp of approval from the Foundation for the Accreditation of Cellular Therapy, a U.S.-based group that vets hospitals.

Patients' cells collected in those hospitals will be flown to a manufacturing facility in the United States or Europe. There, scientists will use the gene-editing system CRISPR-Cas9 to turn off a key gene. This is the first approved therapy to use CRISPR.



At age 34, Nkem Azinge, a government project manager in Abuja, Nigeria, knows that she is within a decade of the average life expectancy of a Nigerian sickle cell patient, and that each pain crisis she has does further damage to her organs. Taiwo Aina for The New York Times

That complex and expensive process is worlds away from the level of care available to most African sickle cell patients today.

Few African countries have even routine screening of newborns for sickle cell, which is standard in wealthy countries. Undiagnosed, those children miss out on a crucial treatment with penicillin that can stave off the pneumonia that often kills sickle cell patients as babies. In Nigeria, there are estimates that up to half of children with sickle cell die before their fifth birthday.

Beyond that, many do not have access to a medication called hydroxyurea that keeps red blood cells round and reduces episodes of severe pain. That medication costs about $7 per patient per month in Nigeria, which still keeps it beyond the

range of many families, Dr. Nnodu said. Many of her patients struggle to afford even basic pain medications such as folic acid and analgesics, she said.

Until the new gene therapies, the only cure for sickle cell disease was a bone-marrow transplant, in which patients have their stem cells wiped out and replaced by healthy cells from a donor who does not have sickle cell. The procedure is reserved for only the most severe cases because it does not always work and is risky; it kills 5 to 20 percent of those who undergo it, depending on age.

Several medical centers in sub-Saharan Africa have recently started doing bone-marrow transplants for sickle cell patients, but only a handful of wealthy African patients can afford them.

In Dodoma, the capital of Tanzania, Benjamin Mkapa Hospital has done five transplants so far, with the government picking up the bill of about $50,000 per patient. Dr. Stella Malangahe, a hematologist there, said her patients often ask her when the hospital will start offering gene therapy. She has no answer.

A small but growing number of African patients is traveling to India for bone-marrow transplants, where they are cheaper and hospitals have more experience doing them. Fortis Memorial Research Institute in Gurugram, on the edge of New Delhi, has performed such transplants in almost 100 African sickle cell patients, according to Dr. Vikas Dua, the head of pediatric hematology.

Ms. Mgaraganza, the Tanzanian mother who works in a bank in Dar es Salaam, temporarily moved her family to India in September so that her two children could get transplants there. She donated her healthy stem cells, and the children underwent chemotherapy and were infused with the new cells. The final bill for two transplants could be $80,000 — and she is still figuring out how to come up with the money.

The children are battling side effects from the transplant drugs. Ms. Mgaraganza would have preferred they get gene therapy, which she learned about by watching YouTube videos. But her children could not wait until someone finds a way to bring it to Africa, she said.



Ramadhani Gwoto, 12, with his mother, Shani Mgaraganza, took steps while under observation following a bone marrow transfusion in Gurugram, India. Saumya Khandelwal for The New York Times



Ms. Azinge has been debating whether to put a plan to get a bone-marrow transplant in India on hold and try instead to find a way to get Casgevy abroad. Taiwo Aina for The New York Times

Nkem Azinge, a government project manager in Abuja, Nigeria, has been saving money to pay for a bone-marrow transplant in India. At 34, she knows that she is within a decade of the average life expectancy of a Nigerian sickle cell patient, and that each pain crisis she has does further damage to her organs. Now, she is debating whether to put her India plan on hold, and try instead to find a way to get a gene therapy abroad.

“If given the chance at this, I would do it because I am living with unimaginable pain,” she said.

Beyond industry, other research groups including the National Institutes of Health and Dr. Doudna’s Innovative Genomics Institute at Berkeley are developing techniques that would make a gene therapy for sickle cell cheaper and easier to

administer. But experts caution that those approaches have not yet been shown to work and are still many years away.

The population of patients in Africa with sickle cell will continue to grow as screening and access to basic interventions expand and fewer children die as infants. That will increase the odds that two people with the sickle cell genetic trait will have a child who could have the inherited disease.

Dr. Léon Tshilolo treats patients with sickle cell at a hospital in Kinshasa, the capital of the Democratic Republic of Congo. Some struggle to raise even $7 each month for basic medications; two have traveled abroad for bone-marrow transplants. Increasingly, they are wondering about gene therapies.

“My young patients, adolescents, they have the internet,” he said, “and they say, ‘Doctor, I saw that some people who have this are being completely cured — when will this come to Kinshasa?’”

Rebecca Robbins is a reporter covering the pharmaceutical industry. She has been reporting on health and medicine since 2015.

Stephanie Nolen covers global health. She has reported on public health, economic development and humanitarian crises from more than 80 countries around the world.

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